AI FOR PMS · IIIT HYDERABAD · CAPSTONE KICKOFF

The Capstone Project

Teams of 4. Build an AI feature, run evals on golden datasets with human and LLM judges, improve it, take it to real customers, and iterate.

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90 minutes: 30 briefing on the project and deliverables, then 4 hands-on activities of 15 minutes each.

WHERE WE ARE

Nine sessions of concepts. The rest is building.

COVERED SO FAR

AI landscape

AI vs ML vs DL; the PM in the AI stack. Zepto, Sarvam AI.

Uncertainty

Deterministic vs ML product logic; uncertainty.

AI decisions

Build vs rules; scoping ML problems; accuracy bars.

GenAI costs

Tokens, unit costs, hallucination risk, model choice.

Evals & data

Rubrics, assertion tests, ship gates, labelling.

REMAINING SESSIONS → YOUR CAPSTONE: build, evaluate, ship to customers, iterate.

Until now you evaluated other people's AI products. From today, you build your own.

TODAY'S 90 MINUTES

Agenda: brief, then build

TIME	BLOCK	MODE
0:00 - 0:05	Why a capstone, and the course-to-project arc	Talk
0:05 - 0:30	The project: brief, evals, customers, deliverables, milestones, grading	Talk
0:30 - 0:45	Activity 1 - Form your team of 4 and pick the problem	Hands-on
0:45 - 1:00	Activity 2 - Scope it: build vs rules + AI PRD skeleton	Hands-on
1:00 - 1:15	Activity 3 - Golden dataset + eval rubric v0	Hands-on
1:15 - 1:30	Activity 4 - Pitch + customer plan	Hands-on

Laptops open for every activity. Each activity ends with an artifact your team keeps - nothing is throwaway.

WHY A CAPSTONE

Reading about evals is not running evals

What changes today

Concepts become artifacts.

Opinions become eval scores.

Classmates become teammates.

Users become real customers.

And 'I think it works' becomes a number you defended.

Course outcomes you now prove

CO-2 Pick an AI-first problem and justify it over rules.

CO-3 Produce real PM deliverables: AI PRD, rubrics, metrics.

CO-4 Design and defend an evaluation framework with a ship gate.

CO-5 Exercise judgment when the model is wrong or unfair.

The capstone is the course. Everything from the nine sessions shows up again - now with your name on it.

THE CAPSTONE LOOP

Build, evaluate, improve, take to customers, iterate



A working AI feature on real inputs

Golden dataset + human review + LLM judge

Fix the top failure class, re-run evals

Real users try it; capture verbatims

Repeat until evals and users hold up

Use the vernacular order-support assistant from earlier sessions as a reference. Your team now runs the full loop on its own product. Aim for at least three full iterations.

TEAMS

Form a team of 4 - locked in during Activity 1

Why four

Enough hands for a golden dataset, two kinds of evals and five customer calls.

Small enough that every person owns a workstream and can defend every decision.

Choose people you can argue productively with, not just friends.

Suggested roles (rotate each week)

PM lead - scope, PRD, priorities

Data & evals - golden set, rubrics, judges

Prompt & model ops - prototype, fixes, runs

Customer & UX - interviews, failure states, copy

Roles rotate. Everyone owns the full project and can be asked to explain any part.

THE BRIEF

A real AI feature for a real problem

Pick a problem worth AI. Apply the build-vs-rules test from Unit 3: if a rules engine solves it, pick again.

Target users you can actually reach this month - a campus community, a family business, a local service, your own circle.

Ship a working prototype. LLM APIs, no-code tools and notebooks are all fine; this is PM-quality engineering, judged on product thinking, not code.

It must run end-to-end on real inputs, including the messy ones.

Small and real beats big and imaginary. A vernacular order-support assistant for tier-2 shoppers beats a vague 'AI for everything'.

EVALS, PART 1

Evals start with a golden dataset

What it is

30-50 real input-output pairs your team writes and labels by hand.

Mix: happy paths, edge cases, adversarial inputs.

Each pair carries the correct answer and why it is correct.

How you use it

Source inputs from reviews, support tickets, interviews - real language, not what you hope users say.

Every eval run scores against this set. Version it; never edit it to make a run look better.

It is the scoreboard for every iteration.

Define your minimum acceptable performance before launch - the ship gate from Unit 5. No gate, no launch.

EVALS, PART 2

Two judges: humans and LLMs

Human evaluation

A rubric with 3-5 criteria: accuracy, groundedness, tone, safety, task success.

Two or more raters per sample; log every disagreement.

Slow and expensive - spend it on the outputs that matter most.

LLM-as-judge

Assertion tests plus rubric prompts that run on every change.

Track pass rate per golden-set version; watch precision and recall where they apply.

Spot-check the judge against human ratings - an unverified judge is just another model.

The rule from Session 3 still holds: no AI output is used until it is verified. That now includes your LLM judge.

IMPROVE

Failure analysis is the real work

1. Run the full golden set through the current build
2. Cluster the failures - what kind of wrong is it?
3. Pick the single worst failure class to fix
4. Fix it: prompt, data, guardrail, or UX
5. Re-run evals. Log the delta. Repeat.

Your iteration log - every run, every failure class, every fix, every delta - is itself a graded deliverable.

TAKE IT TO CUSTOMERS

Real users, not classmates

Five or more real users from your target segment try the feature. Classmates and teammates do not count.

Watch them use it. Do not explain. Capture verbatims and where they get stuck.

A failure in a real user's hands teaches more than ten failures in a demo.

Design for the failure states: disclose that it is AI, show uncertainty honestly, and have a graceful fallback - a preview of the AI-in-UX and ethics unit.

Customer evidence is quoted, not summarised from memory. Verbatims and counts beat vibes.

EXPECTATIONS

End-of-course deliverables

#	DELIVERABLE (ONE PER TEAM)
1	One-page AI PRD: problem, users, build-vs-rules case, model approach, risks
2	Golden dataset: 30-50 labelled input-output pairs, versioned
3	Eval framework: human rubric + LLM judge + your ship gate
4	Eval results across at least three iterations, with the failure analysis
5	Customer evidence: 5+ real-user sessions, verbatims, what changed
6	Final demo and a 10-minute presentation in the last class

Everything is a group submission. Keep artifacts in one shared folder from day one.

MILESTONES

How the remaining sessions map to the loop

WHEN	WHERE YOU SHOULD BE
Today	Team of 4 locked, problem picked, PRD skeleton drafted
Next session	Prototype v0 running + golden dataset drafted (first 20 pairs)
Middle sessions	First full eval run; ship gate defined; failure analysis started
Later sessions	Customer sessions done; two or more improvement iterations logged
Final class	Demo + 10-minute presentation + all six deliverables submitted

Dates follow the class calendar (Wed + Sat sessions). Each session's concepts feed the milestone it sits next to.

HOW IT IS EVALUATED

Grading rewards evidence, not theatre

DIMENSION	WHAT 'GOOD' LOOKS LIKE	WEIGHT*
Problem & PRD	AI-first problem, honest build-vs-rules case, clear users	**15%**
Eval rigour	Real golden set, working rubric + LLM judge, defensible ship gate	**30%**
Iteration	Three or more logged improvement cycles driven by failures	**20%**
Customer learning	5+ real users; verbatims; visible changes from feedback	**15%**
Demo & presentation	Working feature, clear story, honest about limits	**20%**

* Indicative weighting - confirm in class. Evals and iteration outweigh the demo.

ACTIVITY 1 · 15 MINUTES

0:30 - 0:45

Form your team and pick the problem

0-5 min	Form your team of 4. Exchange numbers, create the group chat, claim a row on the class sheet.
5-12 min	Each member pitches one problem in 60 seconds. Vote. Then stress-test the winner: does it fail the rules test? Is the error tolerance workable?
12-15 min	Write a one-line problem statement: who hurts, what hurts, why now.

OUT THE DOOR WITH: team name, 4 members, one-line problem statement on the shared sheet.

ACTIVITY 2 · 15 MINUTES

0:45 - 1:00

Scope it: build vs rules + PRD skeleton

0-3 min	Name the user and their job-to-be-done in one sentence each.
3-8 min	Write the build-vs-rules case: what accuracy do you need, what do errors cost, why is AI worth it?
8-13 min	Fill the PRD skeleton: problem, users, approach, risks, success metric.
13-15 min	Swap with the nearest team. Give one sharp critique; take one.

OUT THE DOOR WITH: a half-page AI PRD draft your team can defend.

ACTIVITY 3 · 15 MINUTES

1:00 - 1:15

Golden dataset + eval rubric v0

0-5 min	Write 10 golden examples: 5 happy paths, 3 edge cases, 2 adversarial. Label the correct output and why.
5-11 min	Define 3 rubric criteria with pass and fail anchors - concrete, not vibes.
11-15 min	Decide the split: what humans rate, what the LLM judge scores, and how you will spot-check the judge.

OUT THE DOOR WITH: 10 labelled golden examples + a 3-criterion rubric you can run tomorrow.

ACTIVITY 4 · 15 MINUTES

1:15 - 1:30

Pitch it, then find your customers

0-8 min	Pair with another team. Each team gets 90 seconds: the problem, the approach, how you will know it works.
8-13 min	For every pitch: one risk and one suggestion from the room. Write yours down.
13-15 min	Draft your customer plan: name 5 real users you can reach and how you reach them this week.

OUT THE DOOR WITH: a pitch delivered, feedback collected, and a 5-name customer list with dates.

THE BAR

What good looks like - and what fails quietly

Strong projects

Narrow problem, sharply defined users

Evals exist before the first demo

Failures logged, clustered, fixed, re-tested

Real-user evidence with anonymised IDs

Honest about where the model still fails

Quiet failures

A demo-only project that never saw an eval

Rubric written after seeing the results

Golden set with no edge or adversarial cases

Classmates presented as 'customers'

Metrics quoted with no baseline and no gate

BEFORE THE NEXT SESSION

Leave today moving

1. Finalise your one-page AI PRD.
2. Grow the golden dataset to 30-50 labelled pairs.
3. Get prototype v0 running end-to-end on real inputs.
4. Book the first two customer sessions.

Questions anytime - class group or office hours.

You have everything the course taught you. Now build.